Pool Accessible Wheelchair Redesign

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Background

Aquatic therapy is used as a low-impact treatment option for a variety of conditions and impairments.

Many patients at the Bellevue Aquatic Center utilize a pool accessible wheelchair (PAW) to access the pool for therapy.



Wave Therapies' heated pool for aquatic therapy.[2]

Requirements

Wave Therapies needs a wheelchair that can be used to transfer their clients into the pool. The wheelchair must be comfortable, reliable, and easy to replicate.

Current PVC PAW issues include:

- Prone to tipping backwards
- Takes significant time to sink and stabilize
- Metal hardware pokes into patients
- Uncomfortable seat bar
- Requires constant and difficult maintenance
- Awkward position with arm rests of different lengths



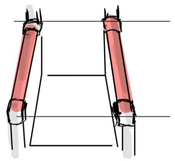
Current PVC PAW

Problem Statement

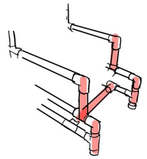
Design a reproducible PAW that makes patients feel secure and comfortable by reducing floating and tipping.

Design and Development

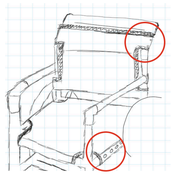
Initial Sketches & CAD



Stability: Changed armrests to be same length as one another



Stability: Added internal ballasts to increase weight and counteract tipping



Comfort: Moved mesh bolts to side



Comfort: Removed horizontal pipe

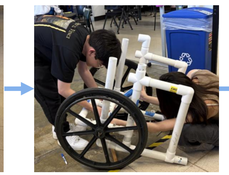


Support: Inserted cross bars

Manufacturing & Materials



PVC Schedule 40 Frame



Attaching airless mag wheels



Coating steel to prevent rust

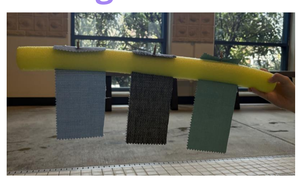


Cementing frame for structure

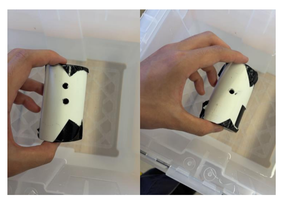


Testing PVC Olefin mesh in the pool

Testing



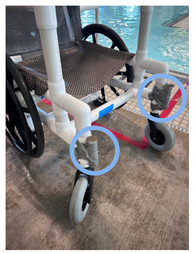
Tested 3 mesh fabrics for chlorine resistance, drainage, and patient comfort



Tested different hole location to determine fastest sinking



Tested different brake mounting systems to determine best outcome



2 x 2.5 lb ballasts



1 x 5 lb ballast



2 x 5 lb ballasts



2 x 5 lb ballasts + 2 x 2.5 lb ballasts



1 x 5 lb ballast + 2 x 2.5 lb ballasts

Tested different steel ballast placements to decrease sinking time and minimize tipping

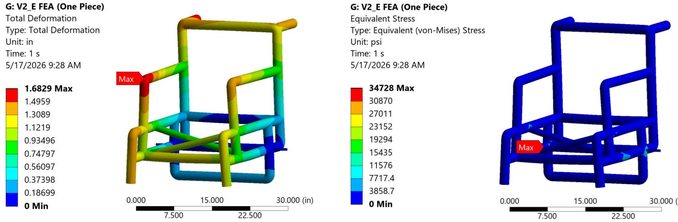
Final Design



Final CAD Design



Final Prototype



Maximum weight capacity: 350 lbs. | Factor of Safety: 1.04

Improvements Made

- Decreased sinking time by approx. 80%
- Greatly reduced chances of tipping backwards
- Inserted mesh side panels to prevent patients from rubbing against wheels
- Made adjustments that allow for easier repair and replacement
- Increased overall comfort and durability

Future Work

- Explore carbon fiber wheellock options
- Special order curved PVC for headrest and footrest
- Design and attach removable footrest

Acknowledgements

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- Krista Torseth
- Wave Therapies
- Bellevue Aquatic Center